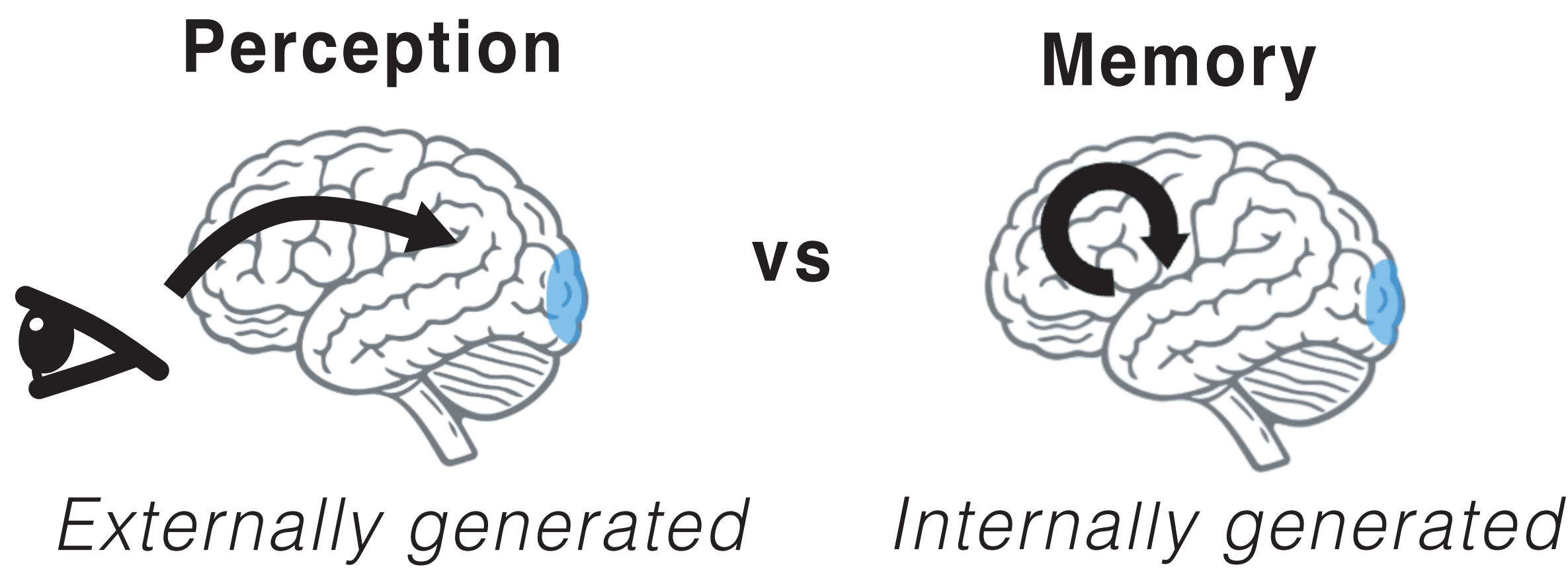


How is neural activity during memory generated in sensory cortex?

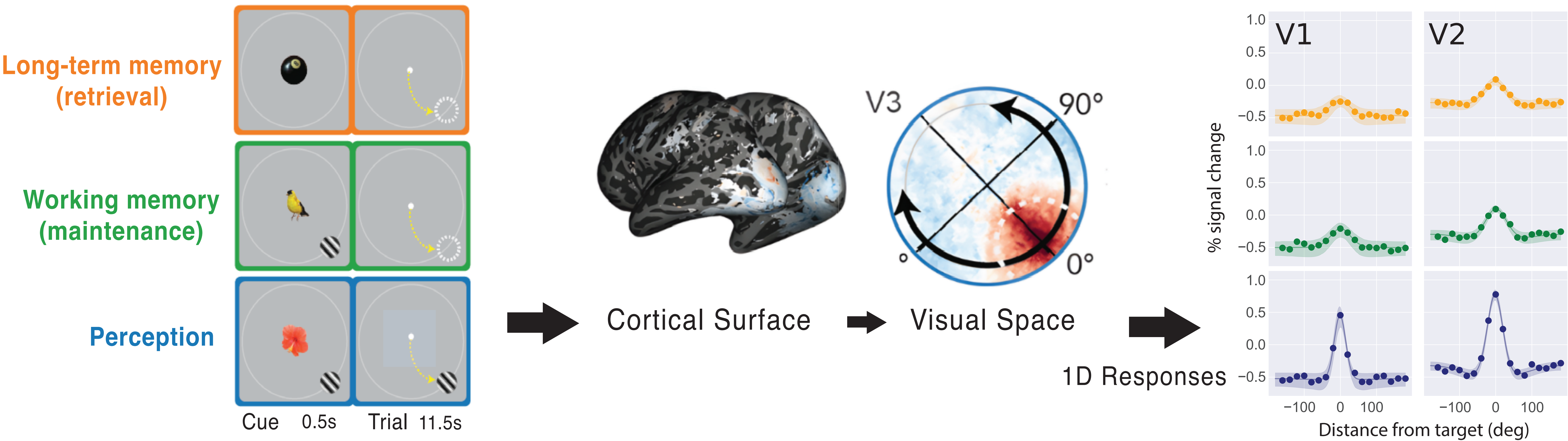
Background: Visual cortex is not only used for perception; it also supports long-term memory (Favila et al., 2022) and working memory (Curtis and Sprague, 2021).

Goal: To understand how memory-related activity in visual cortex arises.

Approach: We asked how activity levels in other brain areas are related to activity patterns in visual cortex during long-term memory and working memory.



Woodry et al., 2025: Spatially tuned memory responses in visual cortex, averaged across trials

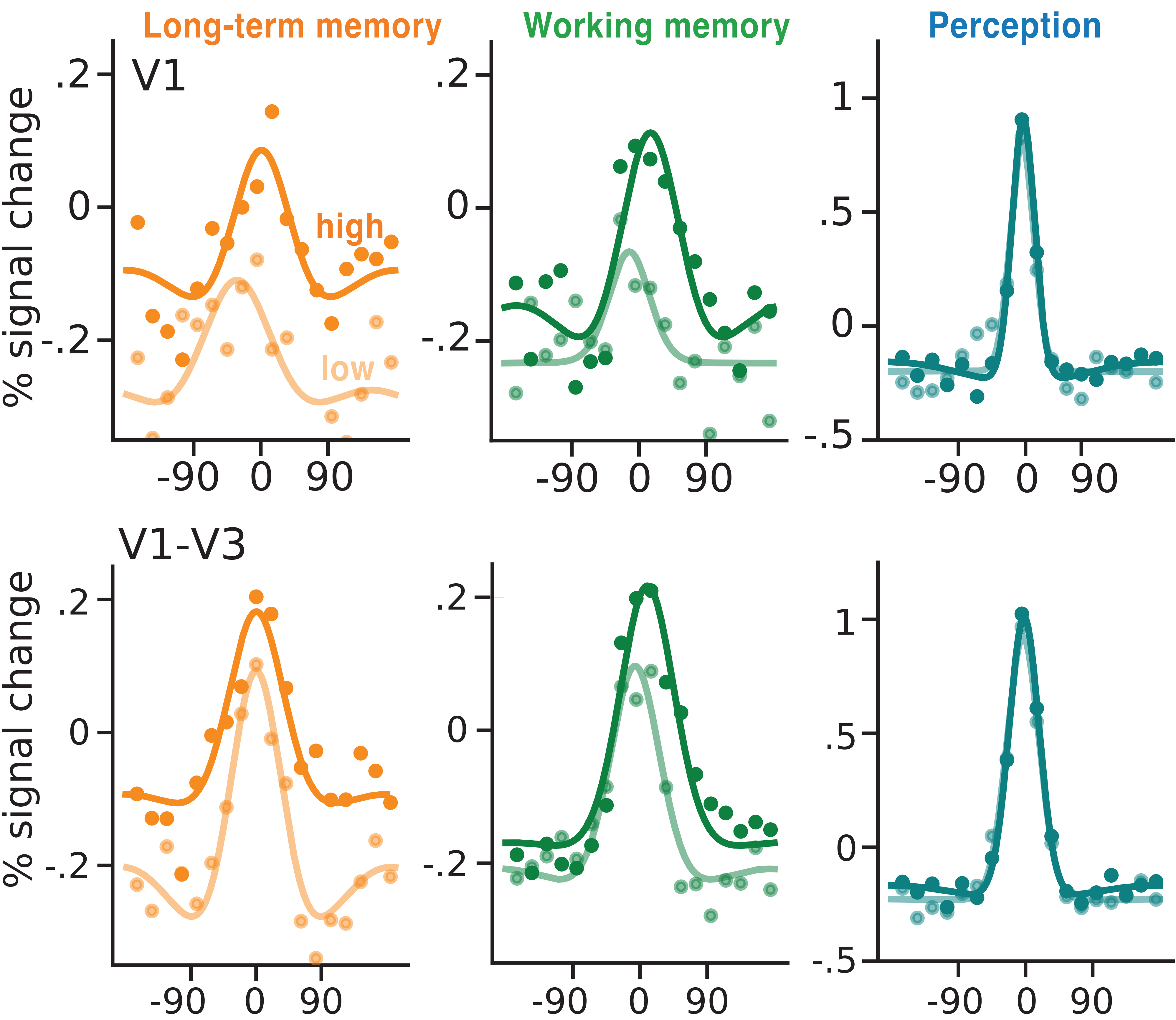


Visual cortex memory responses separated by activity in hippocampus and prefrontal cortex

- We binned trials by fMRI BOLD amplitude in **posterior hippocampus** and **prefrontal cortex (superior precentral sulcus)** for each participant and each task.
- Averaged the top 1/3 plotted in dark colors, bottom 1/3 in light colors.
- Visual cortex responses were averaged from 8 to 15 s post-cue (due to the delayed nature of the hemodynamic response).

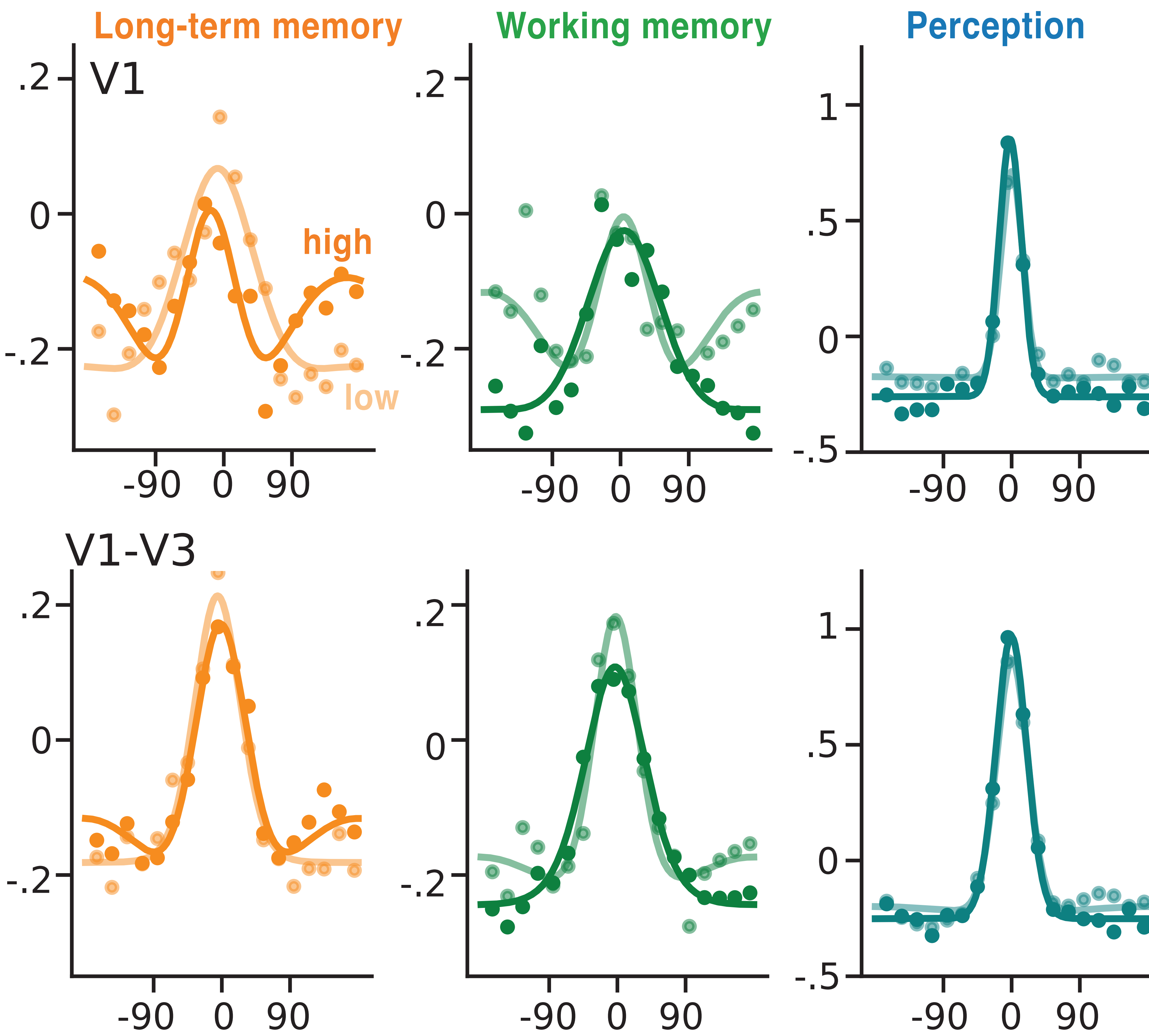
Posterior hippocampus

During long-term memory, visual cortex amplitude is higher when posterior hippocampal activity is higher.



Superior precentral sulcus

During working memory, the visual cortex response is more sharply tuned when superior central sulcus activity is higher.



Hippocampus and prefrontal cortex differentially shape memory responses in early visual cortex

- Hippocampus is more strongly linked to long-term memory responses.
- Prefrontal cortex is differently linked to working memory and long-term memory responses

There are multiple top-down pathways to shape sensory brain activity.

References

Woodry, R., Curtis, C. E., & Winawer, J. (2025). Feedback scales the spatial tuning of cortical responses during both visual working memory and long-term memory. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 45(17).

Favila, S.E., Kuhl, B.A. & Winawer, J. Perception and memory have distinct spatial tuning properties in human visual cortex. *Nat Commun* 13, 5864 (2022)

Curtis CE, Sprague TC. Persistent Activity During Working Memory From Front to Back. *Front Neural Circuits*. 2021 Jul 21;15:696060